

IEOR 4101: Probability, Statistics and Simulation

Lecture 11: Maximum Likelihood Estimation

General Motivation for Statistical Inference

- We collect data about some characteristics of a population
- The entire population may be very big (think about all the people in US etc.), hence observing everyone in the population is impossible, and we can only collect data from a subset of the population
- The characteristic you are interested in knowing about is called the *parameter* (more details later)
- From the data, you calculate a *statistic* that hopefully closely approximates the parameter

General Motivation for Statistical Inference

Common (and sometimes reasonable) assumptions about the population and the data collection process:

- Each individual in the population has a characteristic that is randomly drawn from the same probability distribution and they are all independent
- The data are collected from a random sampling from the population such that, each data point is a random realization from the probability distribution governing the characteristic of the population

Statistical Inference

- Definition: If $X_1, X_2, \dots, X_n$ are independent random variables following the same probability distribution F , then they constitute a (random) sample from the distribution F
- F is unknown (controlling the characteristic of the population)
- Goal of statistics: infer knowledge of F based on the sample

Statistic and Estimator

- Definition: a *statistic* is a number calculated from the sampled data

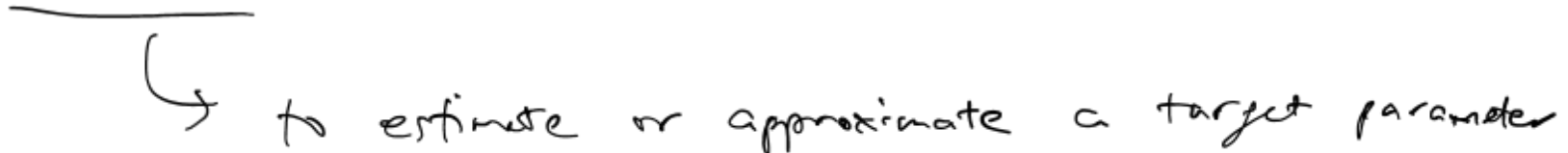
- E.g., Sample mean:

$$\bar{X} = (X_1 + X_2 + \dots + X_n)/n$$

- E.g., Sample variance:

$$S^2 = [(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \dots + (X_n - \bar{X})^2]/(n-1)$$

- A statistic that is used to approximate the parameter is called an *estimator*

 to estimate or approximate a target parameter

Statistical Inference

- F may be completely unknown, or may be specified up to some unknown parameters
 - If F is completely unknown – *nonparametric* inference
 - If F is specified up to a set of unknown parameters – *parametric* inference

E.g., F is specified as a normal distribution function with unknown mean and variance. Nonparametric or parametric? *Parametric*

- We focus on parametric inference, so there is a target parameter that we want to estimate

Parameter Estimation

Examples:

- Poisson distribution has one parameter λ
- Binomial distribution has two parameters n and p
- Normal distribution has two parameters μ and σ^2

Parameter Estimation

- Consider the case that F is a normal distribution with unknown μ and σ^2
- We have seen that a reasonable estimate for the population mean μ is the sample mean $\bar{X} = (X_1 + X_2 + \dots + X_n)/n$. Why?

Yes.

? {

- $E[\bar{X}] = \mu$
- addition of normals = normal
- CLT if $n \geq 3$.

Yes because

$$\bar{X} \approx \mu$$

when n is large, by LLN.

Parameter Estimation

- Example. $X_1, X_2, \dots, X_n$ are a random sample obtained from the distribution $\text{Unif}(0, \theta)$, θ unknown
- Wish to obtain an estimate of θ

Parameter Estimation

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- Wish to obtain an estimate of θ
- Idea 1: Note the expectation is $\theta/2$, so use $(X_1 + X_2 + \dots + X_n)/n$ to approximate $\theta/2$

$$\text{estimator} = 2\bar{X}$$

This estimator is good by LLN.

Parameter Estimation

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- Wish to obtain an estimate of θ
- Idea 1: Note the expectation is $\theta/2$, so use $(X_1 + X_2 + \dots + X_n)/n$ to approximate $\theta/2$
- Idea 2: Use $\max\{X_1, X_2, \dots, X_n\}$ to approximate θ

Parameter Estimation

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- Wish to obtain an estimate of θ
- Idea 1: Note the expectation is $\theta/2$, so use $(X_1 + X_2 + \dots + X_n)/n$ to approximate $\theta/2$
- Idea 2: Use $\max\{X_1, X_2, \dots, X_n\}$ to approximate θ
- The second idea using $\max\{X_1, X_2, \dots, X_n\}$ is called the *maximum likelihood estimator* and is arguably a better estimator for this problem

Maximum Likelihood Estimation

- Let $X_1, X_2, \dots, X_n$ be a random sample from distribution F_θ , with density function $f(x;\theta)$
- Consider the joint density function that generates the whole set of sample (assuming independence)
$$g(x_1, x_2, \dots, x_n; \theta) = f(x_1; \theta)f(x_2; \theta) \dots f(x_n; \theta)$$
- Given observations $X_1, X_2, \dots, X_n$, $g(X_1, X_2, \dots, X_n; \theta)$ is a function of θ .
- This function is called the *likelihood function*, which tracks the “likelihood” of the observations being the realized values. Intuitively, the higher is this function value, the more likely is the parameter being the correct one
- Find $\hat{\theta}$ that maximizes the function $g(X_1, X_2, \dots, X_n; \theta)$
- This is the same as finding $\hat{\theta}$ that maximizes $\log[g(X_1, X_2, \dots, X_n; \theta)]$, the *log-likelihood function*, which is often easier
- The estimator $\hat{\theta}$ is the *maximum likelihood estimator*

For iid data $x_1, \dots, x_n$,

joint pdf of $x_1, \dots, x_n$ $g(x_1, \dots, x_n; \theta)$

$$= f(x_1, \theta) f(x_2, \theta) \dots f(x_n, \theta)$$

because $x_1, \dots, x_n$ are indep and each of them has the same pdf $f(x, \theta)$.

The likelihood function $g(x_1, \dots, x_n; \theta)$ is a function of θ , and maximum Likelihood estimator is the maximizer of $g(x_1, \dots, x_n; \theta)$ over θ .

Maximum Likelihood Estimation

- Example. Consider $X_1, X_2, \dots, X_n$ from the distribution $\text{Unif}(0, \theta)$, θ unknown

- Density function $f_i(x; \theta)$ of X_i is

$$f_i(x \mid \theta) = \begin{cases} \frac{1}{\theta} & \text{if } 0 < x < \theta, \\ 0 & \text{otherwise} \end{cases}$$

- Joint density function of $(X_1, X_2, \dots, X_n)$ is $g(x_1, x_2, \dots, x_n; \theta) = f_1(x_1; \theta)f_2(x_2; \theta)\dots f_n(x_n; \theta)$ (by independence)

$$g(x_1, x_2, \dots, x_n \mid \theta) = \begin{cases} \frac{1}{\theta^n} & \text{if } 0 < x_1, \dots, x_n < \theta, \\ 0 & \text{otherwise} \end{cases}$$

$$X_1, \dots, X_n \sim \text{Unif}(0, \theta)$$

$$\text{pdf of } X_i = \frac{1}{\theta} \quad \text{if } 0 < x_i < \theta \quad \text{and } = 0 \quad \text{otherwise}$$

$$\text{joint pdf of } X_1, \dots, X_n = \frac{1}{\theta^n} \quad \text{if } \begin{array}{c} 0 < x_1 < \theta \\ \vdots \\ 0 < x_n < \theta \end{array}$$

$$g(x_1, \dots, x_n; \theta) = \frac{1}{\theta^n} \quad \rightarrow \quad \text{and } = 0 \quad \text{otherwise}$$

$$\text{MLE } \hat{\theta} \text{ maximizes } g(x_1, \dots, x_n; \theta)$$

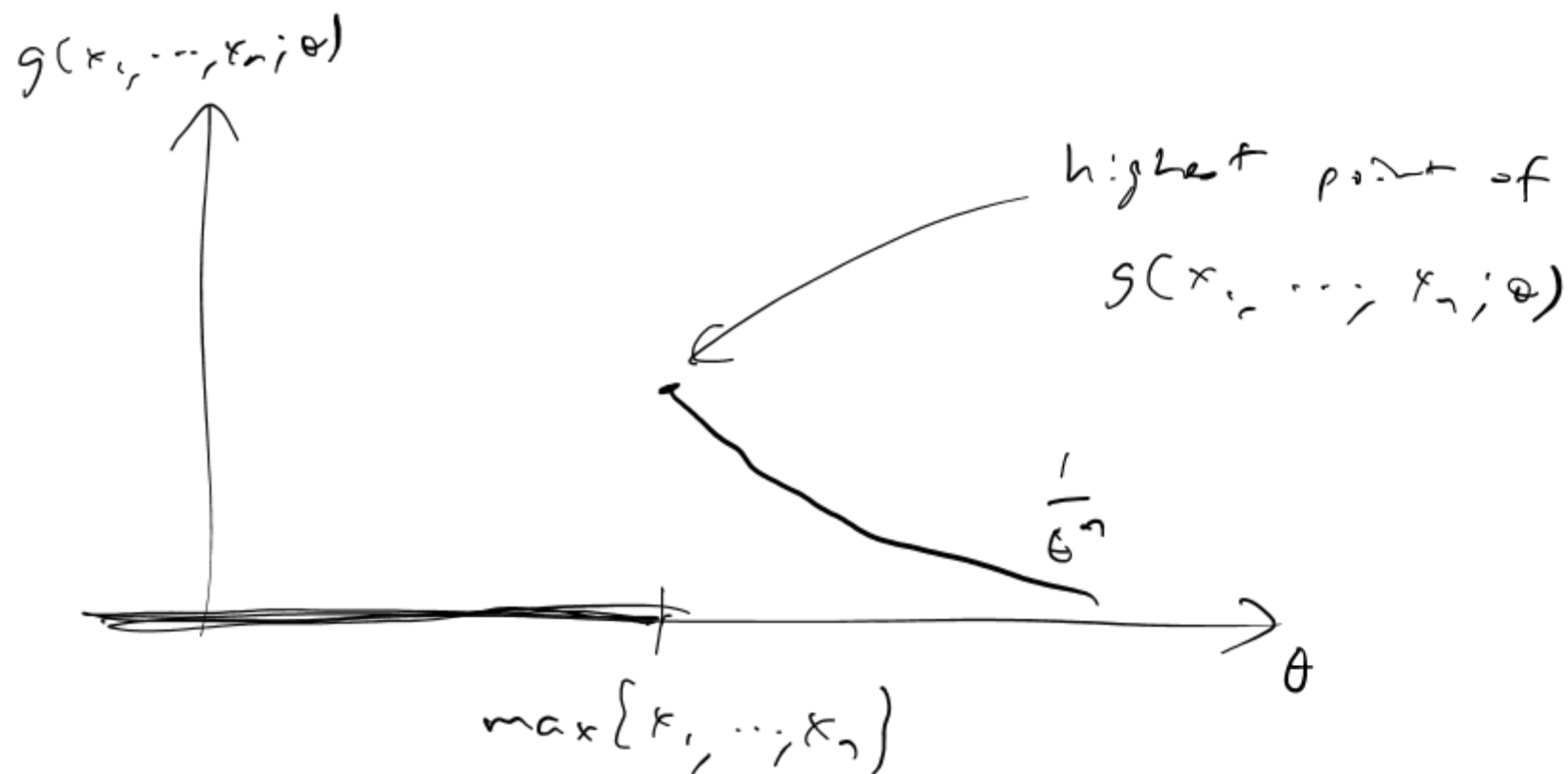
That is,

$$g(x_1, \dots, x_n; \theta) = \frac{1}{\theta^n} \quad \text{if} \quad \left. \begin{array}{l} 0 < x_1 < \theta \\ \vdots \\ 0 < x_n < \theta \end{array} \right\} \begin{array}{l} \min\{x_1, \dots, x_n\} > 0 \\ \max\{x_1, \dots, x_n\} < \theta \end{array}$$

and 0 otherwise

$$\Rightarrow g(x_1, \dots, x_n; \theta) = \frac{1}{\theta^n} \quad \text{if} \quad \theta > \max\{x_1, \dots, x_n\}$$

and 0 otherwise



$$\hat{\theta} = \max\{x_1, \dots, x_n\}.$$

Maximum Likelihood Estimation

- The smaller the θ , the larger $1/\theta^n$ is
- To maximize $1/\theta^n$, we minimize θ
- But the minimal value θ can take is $\max\{X_1, X_2, \dots, X_n\}$
- Thus, $\hat{\theta} = \max\{X_1, X_2, \dots, X_n\}$ is the maximum likelihood estimator of θ

Maximum Likelihood Estimation

- **Example (Estimation of Bernoulli parameter).** We perform n independent Bernoulli trials, each with parameter p , where p is unknown. What is the maximum likelihood estimator of p ?

$$X_1, \dots, X_n \sim \text{Ber}(p)$$

$$\text{MLE } \hat{p} = ?$$

$$\text{pmf of } X_i = \begin{cases} p & \text{if } x_i = 1 \\ 1-p & \text{if } x_i = 0 \end{cases}$$

$$\text{Joint pmf of } x_1, \dots, x_n : g(x_1, \dots, x_n; p)$$

$$= \begin{cases} p & \text{if } x_1 = 1 \\ 1-p & \text{if } x_1 = 0 \end{cases} \times \begin{cases} p & \text{if } x_2 = 1 \\ 1-p & \text{if } x_2 = 0 \end{cases} \times \dots \times \begin{cases} p & \text{if } x_n = 1 \\ 1-p & \text{if } x_n = 0 \end{cases}$$

$$\text{MLE } \hat{p} = \max_p g(x_1, \dots, x_n; p)$$

To rewrite it,

$$\text{pmf of } x_i = p^{x_i} (1-p)^{1-x_i}$$

$$\text{joint pmf: } g(x_1, \dots, x_n; p)$$

$$= p^{x_1} (1-p)^{1-x_1} \dots p^{x_n} (1-p)^{1-x_n}$$

$$= p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}$$

likelihood function:

$$g(x_1, \dots, x_n | p) = p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}$$

log-likelihood function:

$$\log g(x_1, \dots, x_n | p) = \sum_{i=1}^n x_i \log p + (n - \sum_{i=1}^n x_i) \log(1-p)$$

Take derivative, and set to 0,

$$\frac{\sum_{i=1}^n x_i}{p} - \frac{n - \sum_{i=1}^n x_i}{1-p} = 0$$

$$\frac{\sum_{i=1}^n x_i}{p} = \frac{n - \sum_{i=1}^n x_i}{1-p}$$

$$\Rightarrow \hat{p} = \frac{\sum_{i=1}^n x_i}{n}$$



Need to check the function is concave,
i.e. 2nd order derivative is negative. (we skip this
step).

Maximum Likelihood Estimation

- **Example (Estimation of Bernoulli parameter).** We perform n independent Bernoulli trials, each with parameter p , where p is unknown. What is the maximum likelihood estimator of p ?
- **Solution.** Data consist of $X_1, X_2, \dots, X_n$ where $X_i = 1$ if the i^{th} trial is a success, and 0 if a failure
- $P(X_i = 1) = p, P(X_i = 0) = 1-p$
- Joint probability mass function is

$$\begin{aligned} & g(x_1, x_2, \dots, x_n \mid p) \\ &= p^{\# \text{ of successes}} \cdot (1 - p)^{\# \text{ of failures}} \\ &= p^{\sum_{i=1}^n x_i} \cdot (1 - p)^{n - \sum_{i=1}^n x_i} \end{aligned}$$

Maximum Likelihood Estimation

- Log-likelihood is

$$\log g(x_1, x_2, \dots, x_n \mid p) = \left(\sum_{i=1}^n x_i \right) \log p + \left(n - \sum_{i=1}^n x_i \right) \log(1 - p)$$

- Differentiate log-likelihood to find $\hat{p}$

$$\begin{aligned} & \frac{d}{dp} \log g(x_1, x_2, \dots, x_n \mid p) \\ &= \frac{d}{dp} \left[\left(\sum_{i=1}^n x_i \right) \log p + \left(n - \sum_{i=1}^n x_i \right) \log(1 - p) \right] \\ &= \frac{1}{p} \sum_{i=1}^n x_i - \frac{1}{1 - p} \left(n - \sum_{i=1}^n x_i \right) \end{aligned}$$

Maximum Likelihood Estimation

- Setting the derivative to 0, we get

$$\frac{1}{\hat{p}} \sum_{i=1}^n x_i - \frac{1}{1 - \hat{p}} \left(n - \sum_{i=1}^n x_i \right) = 0$$

- Algebraic manipulation: $\hat{p} = (x_1 + x_2 + \dots + x_n)/n$
- So the maximum likelihood estimator is

$$\hat{p} = \frac{1}{n} (X_1 + X_2 + \dots + X_n)$$

Maximum Likelihood Estimation

- Example (Estimation of Normal parameters).** $X_1, X_2, \dots, X_n$ are a sample from a normal distribution with unknown μ and σ^2 . Find the maximum likelihood estimators of μ and σ^2

$$\text{pdf of } x_i = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

$$\text{joint pdf of } x_1, \dots, x_n = g(x_1, \dots, x_n; \mu, \sigma)$$

$$= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_1 - \mu)^2}{2\sigma^2}} \dots \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_n - \mu)^2}{2\sigma^2}}$$

$$= \frac{1}{(2\pi)^{\frac{n}{2}} \sigma^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

log-likelihood function:

$$\log g(x_1, \dots, x_n; \mu, \sigma)$$

$$= -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Take partial derivatives w.r.t. μ and σ ,

and set them to 0,

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0$$

$$\Rightarrow \sum_{i=1}^n x_i - n\mu = 0$$

$$\Rightarrow \mu = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}$$

$$-\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

So MLE:

$$\begin{cases} \hat{\mu} = \frac{\sum_{i=1}^n x_i}{n} \\ \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2 \end{cases}$$

Maximum Likelihood Estimation

- **Example (Estimation of Normal parameters).** $X_1, X_2, \dots, X_n$ are a sample from a normal distribution with unknown μ and σ^2 . Find the maximum likelihood estimators of μ and σ^2

- **Solution.** Individual density function is

$$f(x_i) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu)^2}{\sigma^2}}$$

- Joint density function is

$$\begin{aligned} &g(x_1, x_2, \dots, x_n \mid \mu, \sigma) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu)^2}{\sigma^2}} \\ &= \left(\frac{1}{2\pi}\right)^{n/2} \frac{1}{\sigma^n} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right] \end{aligned}$$

Maximum Likelihood Estimation

- Log-likelihood $g(x_1, x_2, \dots, x_n | \mu, \sigma)$ is
$$-\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$
- Taking partial derivatives with respect to μ and σ gives

$$\frac{\partial}{\partial \mu} g(x_1, \dots, x_n | \mu, \sigma) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

$$\frac{\partial}{\partial \sigma} g(x_1, \dots, x_n | \mu, \sigma) = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (x_i - \mu)^2$$

- Setting partial derivatives to 0, we get

$$\hat{\mu} = \sum_{i=1}^n x_i / n$$
$$\hat{\sigma} = \left[\sum_{i=1}^n (x_i - \hat{\mu})^2 / n \right]^{1/2}$$

Maximum Likelihood Estimation

- Unbiased estimator: given $X_1, X_2, \dots, X_n$ a random sample, and let $f(X_1, X_2, \dots, X_n)$ be an estimator of parameter θ . The estimator is *unbiased* if $E[f(X_1, X_2, \dots, X_n)] = \theta$
 - E.g. $E[\bar{X}] = E[(X_1 + X_2 + \dots + X_n)/n] = E[X] = \mu$; so sample mean is always an unbiased estimator of the population mean
 - E.g., $S^2 = [(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \dots + (X_n - \bar{X})^2]/(n-1)$, and we can show that $E[S^2] = \text{Var}(X)$, which justifies the correction of $(n-1)$ in the sample variance
- Maximum likelihood estimator is statistically *efficient* when the sample size is sufficiently large, i.e., it is approximately unbiased and its variance is the smallest among any possible estimators

Generally,

a measurement of error for
estimator $\hat{\theta}$

$$= \frac{\text{mean squared error of } \hat{\theta}}{\text{MSE}}$$

$$= \left(\underbrace{\text{bias of } \hat{\theta}}_{E[\hat{\theta}] - \theta} \right)^2 + \text{var}(\hat{\theta})$$

When sample size n is large,

MLE has almost 0 bias

and its variance is the smallest

possible among all estimators.

$\Rightarrow$ MLE is the best among all possible estimators.